

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile : Course Content Simplification Agent

Domain of Project : Agentic AI / Generative AI in Education

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Deepak B		deepakbalan26@gmail.com ananthibalan64@gmail.com	+ 91 8015170344

Problem Statement : Course Content Simplification Agent

The Challenge – Educational materials often vary in complexity and are not always accessible to learners with different levels of prior knowledge. Students may struggle to grasp key concepts due to jargon heavy or overly advanced explanations in faculty notes and textbooks. The challenge is to develop an AI powered agent that can intelligently analyse academic content and reframe explanations based on the learner's current proficiency—ranging from beginner to expert. This would support more inclusive learning and personalized education delivery at scale

The Objective - Build an **AI-powered Course Content Simplification Agent** that helps students understand complex academic content by breaking it down into clear, easy-to-read explanations based on their learning level.

The solution should:

- **Content Simplification** — Take difficult textbook chapters, lecture notes, or research papers and rewrite them in simple, understandable language.
- **Multi-Level Learning** — Adjust the explanation based on the student's proficiency — Beginner, Intermediate, Advanced, or Expert — so every learner gets the right depth of content.
- **Structured Output** — Deliver the simplified content in organized sections: a plain explanation, key concepts, important points, a real-world example, and a quick revision summary.
- **Built-in Demo Mode** — Allow students to use the tool even without an IBM Cloud account, using a smart offline educational engine.
- **IBM Granite AI Integration** — Connect to IBM Granite language models through IBM watsonx.ai for high-quality, AI-generated simplifications in live mode.
- **Download & Study** — Let students download the simplified content as study notes in Markdown or plain text format for offline revision.

Proposed Solution :

AI-Powered Course Content Simplification Agent

The proposed solution is an **AI-powered agent** built using **IBM watsonx.ai and IBM Granite Models** that converts complex academic content — like textbook chapters, lecture notes, and research papers — into clear, easy-to-understand explanations based on the student's learning level.

The system supports four proficiency levels — **Beginner, Intermediate, Advanced, and Expert** — and delivers five structured outputs for every simplification:

- **Simple Explanation** — Level-appropriate rewrite of the content
- **Key Concepts** — Most important ideas extracted from the input
- **Important Points** — Critical takeaways to remember
- **Examples** — A real-world analogy for better understanding
- **Quick Revision** — A 1–2 sentence summary for fast review

The solution includes a **built-in Demo Mode** for offline use and switches to **Live IBM Granite Mode** automatically when IBM credentials are provided. The interface is built with **Streamlit**, following a simple four-step workflow — paste content, choose level, simplify, and download study notes.

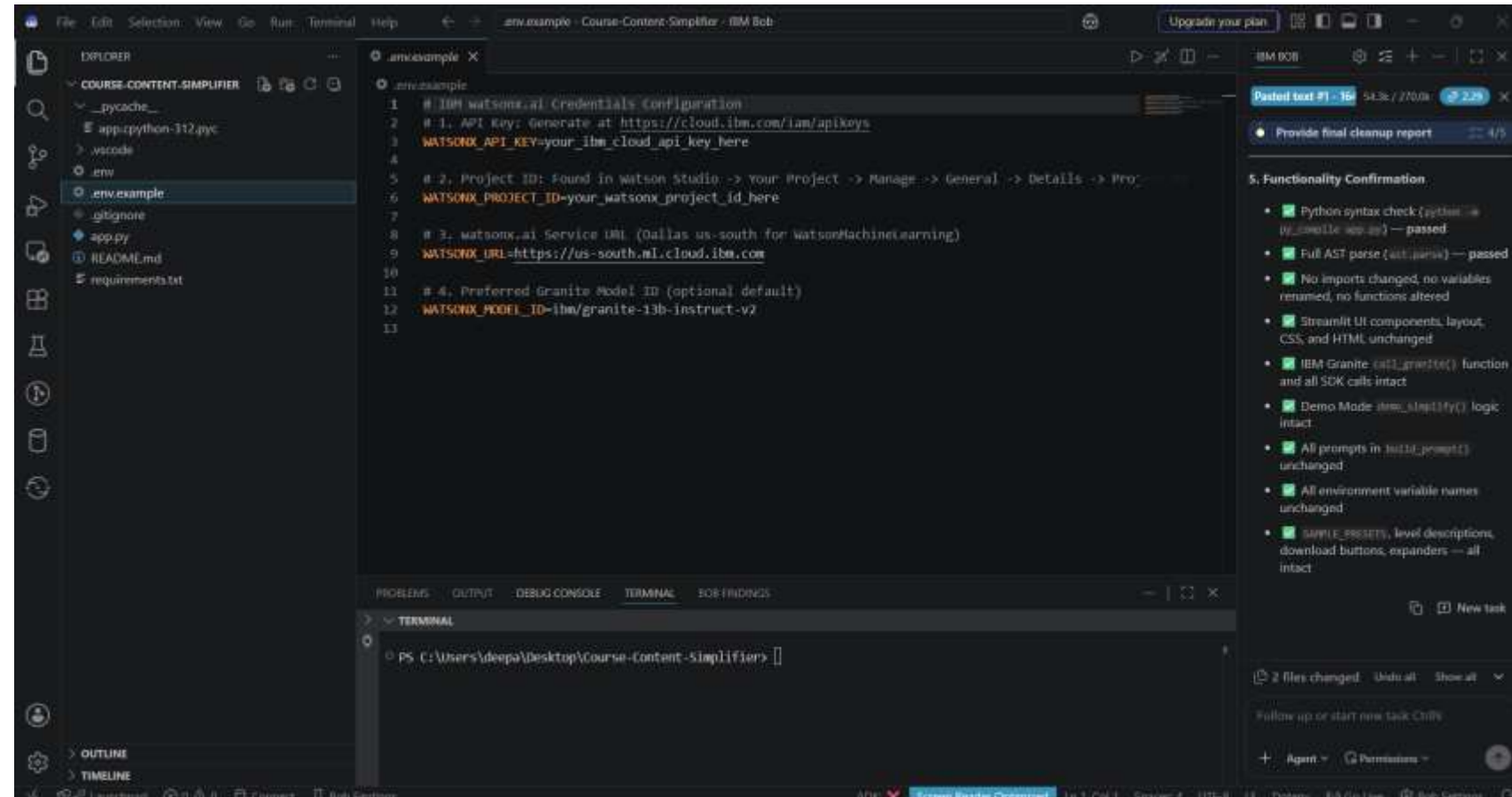
IBM watsonx.ai handles secure model inference while IBM Granite powers the AI reasoning and content generation — making academic learning more **accessible, personalized, and inclusive** for every student.

Technology Used :

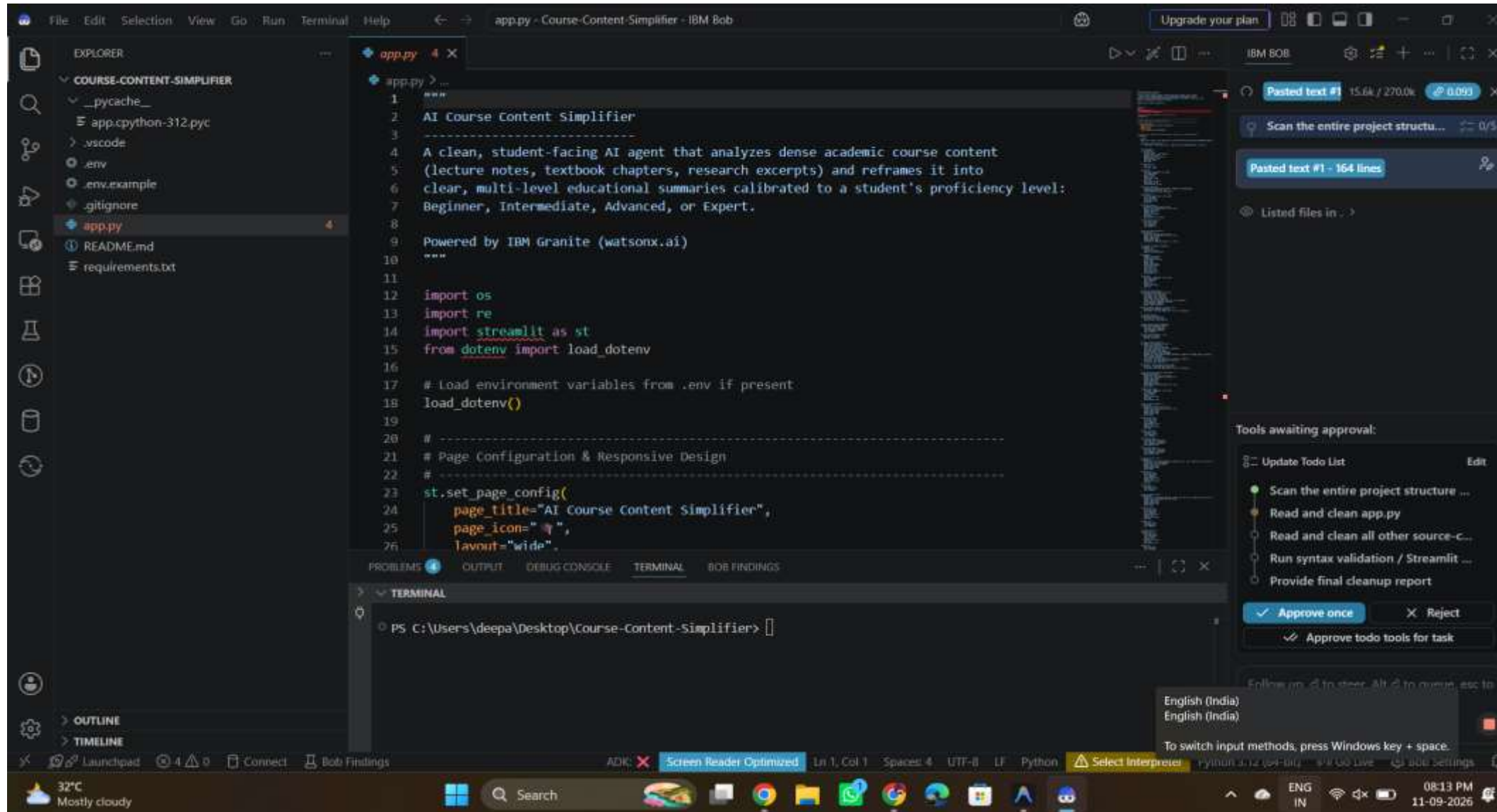
Technology	purpose
Python 3.10+	Core programming language
Streamlit	Web application interface
IBM watsonx.ai	AI model API platform
IBM Granite (ibm/granite-13b-instruct-v2)	AI language model for content simplification
ibm-watsonx-ai SDK	Python library to connect with IBM Granite
python-dotenv	Loads API credentials securely from .env file
IBM Cloud IAM	API key management and authentication
Watson Studio	Project workspace on IBM Cloud

Required Files created using IBM bob -

1. App.py
2. Requirements.txt
3. Style.css
4. .env
5. Index.html



IBM Bob Screenshots:



File Edit Selection View Go Run Terminal Help

README.md - Course-Content-Simplifier - IBM Bob

Upgrade your plan

IBM BOB

EXPLORER

COURSE-CONTENT-SIMPLIFIER

__pycache__

app.cpython-312.pyc

.vscode

settings.json

.env

.env.example

.gitignore

app.py

README.md

requirements.txt

OUTLINE

TIMELINE

README.md

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35

AI Course Content Simplifier

An AI-powered tool that takes complex academic content – like textbook chapters, lecture notes, or research papers – and explains it in simple, easy-to-understand language.

Powered by **IBM Granite** via **IBM watsonx.ai**.

🌟 What It Does

Paste any difficult academic text and choose your learning level. The app will instantly break it down into:

- 📄 **Simple Explanation** – Easy-to-read version of the content
- 🔑 **Key Concepts** – The most important ideas
- 🌟 **Important Points** – Key takeaways to remember
- 💡 **Examples** – A relatable analogy or real-world example
- 📌 **Quick Revision** – A 1-2 sentence summary

🎓 Learning Levels

Level	Best For
🟡 Beginner	Students new to the topic – simple words, everyday examples
🟢 Intermediate	Undergraduate students – clear explanations with basic terms
🔵 Advanced	Senior students – technical depth with proper terminology
🔴 Expert	Researchers – precise, dense, formal language

🚀 How to Run

Step 1 – Install dependencies

89%



Hi, I'm **Bob**

Ask me questions or let me code for you.

Documentation

Recent Tasks

See all

Clean up the entire project by ...

21 min ago ⚡ 2:54

Build Anything. @ for context, / for commands, @ drag to attach

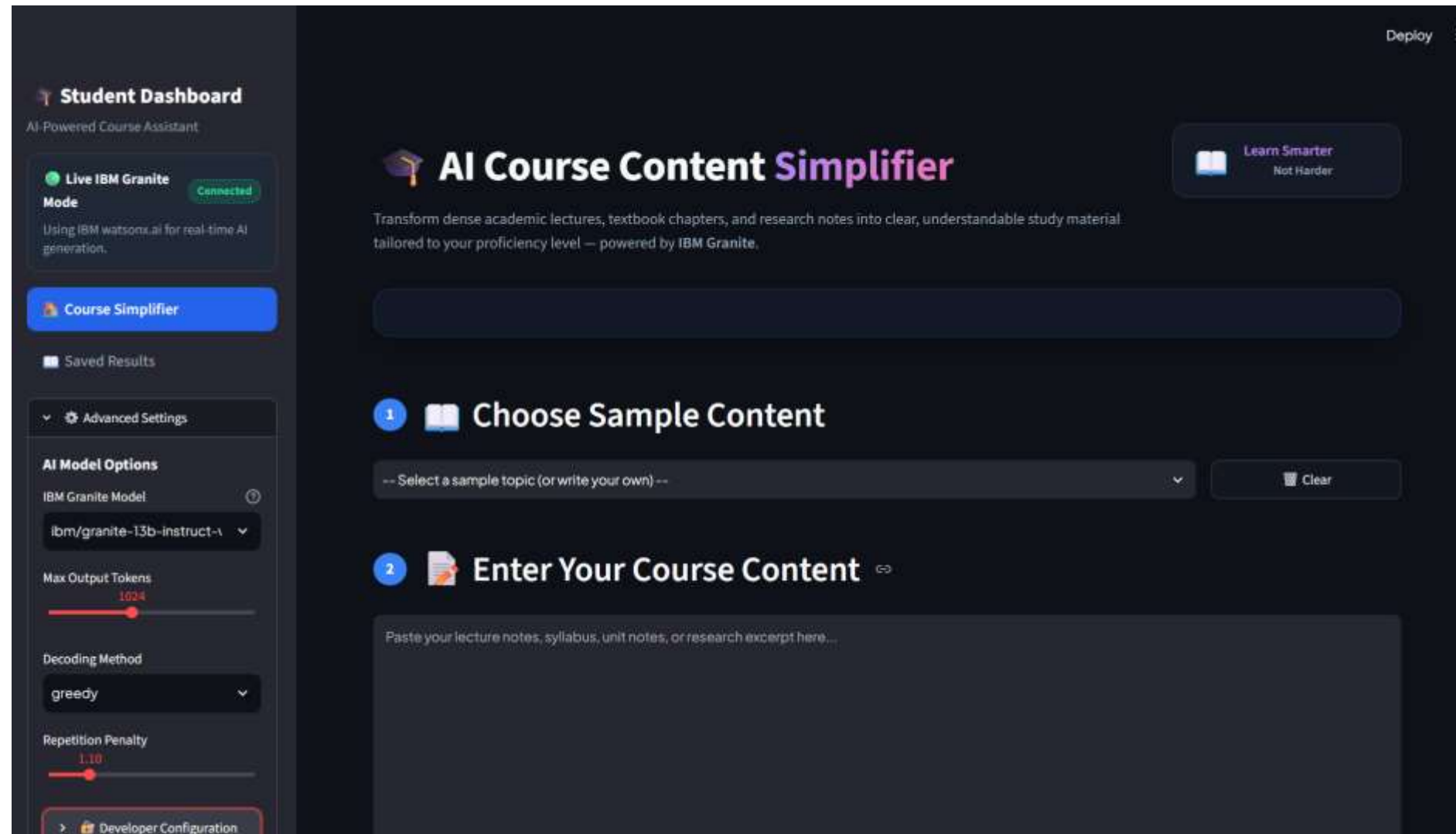
+ Agent ▾ ⚙ Permissions ▾

Role of Agentic AI in the solution

- Analyzes complex academic content provided by students.
- Adapts explanations based on the learner's selected proficiency level.
- Breaks large content into simple, structured sections.
- Generates key concepts, important points, examples, and revision notes.
- Helps students understand and revise course material more efficiently.
- Provides an interactive AI-based learning experience using IBM Granite.



Project Output Screenshot



Project Output Screenshot

The screenshot displays the 'AI Course Content Simplifier' web application. The interface is dark-themed with a sidebar on the left and a main content area on the right.

Sidebar (Left):

- Student Dashboard** (with a graduation cap icon): AI-Powered Course Assistant.
- Mode Toggle:** 'Demo Mode Active' (yellow circle) and 'Offline' (yellow pill).
- Course Simplifier** (blue button with a book icon).
- Saved Results** (document icon).
- Advanced Settings** (gear icon).
- Powered by IBM** (at the bottom).

Main Content Area (Right):

- Header:** 'AI Course Content Simplifier' (with a graduation cap icon) and a 'Deploy' button with a menu icon.
- Tagline:** 'Transform dense academic lectures, textbook chapters, and research notes into clear, understandable study material tailored to your proficiency level — powered by IBM Granite.'
- Input Field:** A large, empty dark blue rounded rectangle for pasting content.
- Step 1: Choose Sample Content** (with a book icon):
 - A dropdown menu showing 'Physics: Quantum Superposition & Wavefunction Collapse'.
 - A 'Clear' button with a trash icon.
- Step 2: Enter Your Course Content** (with a document icon):
 - A text area containing a paragraph about quantum superposition: 'In quantum mechanics, quantum superposition is a fundamental principle stating that any two or more quantum states can be added together ('superposed') and the result will be another valid quantum state. Conversely, every quantum state can be represented as a sum of two or more other distinct states. A particle exists in all possible states simultaneously until it is measured. Upon physical observation, the wavefunction collapses into a definite single state according to the probabilistic Born rule.'

Project Output Screenshot

The screenshot displays a 'Student Dashboard' for an 'AI-Powered Course Assistant'. The dashboard includes a sidebar with navigation options: 'Demo Mode Active' (with an 'Offline' toggle), 'Course Simplifier' (highlighted in blue), 'Saved Results', and 'Advanced Settings'. The main content area is titled 'Simplified Content' and indicates it is 'Tailored for Beginner proficiency - Generated by Educational Engine - 316 words (~2 min read)'. The content is organized into three sections: 'Simple Explanation', 'Key Concepts', and 'Important Points'. The 'Simple Explanation' section provides a basic definition of quantum superposition. The 'Key Concepts' section lists three points: 'Core Process', 'Operational Mechanism', and 'Comprehension Focus'. The 'Important Points' section lists three key takeaways about the systematic conversion of components, regulation of stages, and essential foundational knowledge.

Student Dashboard

AI-Powered Course Assistant

Demo Mode Active

Offline

Using the built-in educational engine.

Course Simplifier

Saved Results

Advanced Settings

Powered by IBM

Deploy

Simplified Content

Tailored for Beginner proficiency - Generated by Educational Engine - 316 words (~2 min read)

Simple Explanation

Simple Explanation In simple everyday terms: In quantum mechanics, quantum superposition is a fundamental principle stating that any two or more quantum states can be added together ('superposed') and the result will be another valid quantum state. Instead of getting confused by complex technical terms, the key idea is that the system takes starting materials and transforms them into a useful form. Conversely, every quantum state can be represented as a sum of two or more other distinct states. This ensures that all essential tasks get the power and resources they need without any wasted effort.

Key Concepts

- **Core Process:** In quantum mechanics, quantum superposition is a fundamental principle stating that any two or more quantum states can be added together ('superposed') and the result will be another valid quantum state.
- **Operational Mechanism:** Conversely, every quantum state can be represented as a sum of two or more other distinct states.
- **Comprehension Focus:** Calibrated specifically for Beginner learners to emphasize cause-and-effect understanding.

Important Points

- The primary driver is the systematic conversion of starting components (In quantum mechanics, quantum superposition is a fundamental principle.).
- Each stage in the pathway is tightly regulated to prevent resource waste and maintain equilibrium.
- Essential foundational knowledge frequently tested in academic exams and applied coursework.

Novelty and Uniqueness

- **Learner-Level Adaptation:** The system adjusts the complexity of explanations based on the student's selected learning level.
- **Structured Simplification:** Instead of simply summarizing, it organizes content into **Simple Explanation, Key Concepts, Important Points, Examples, and Quick Revision**.
- **AI-Based Academic Assistance:** IBM Granite is used to transform technical and lengthy study material into student-friendly content.
- **Single-Input to Multiple Learning Outputs:** One piece of course content is converted into different study formats from the same input.
- **Student-Centered Design:** The interface is designed specifically around common student needs such as understanding difficult topics and quick revision.
- **Demo + Live AI Support:** The system can work in Demo Mode while also supporting live IBM Granite integration when configured.

Git Hub Link

<https://github.com/deepakcodes-2028/Course-Content-Simplification-Agent.git>

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Future Scope

- **Personalized Learning:** Adapt explanations automatically based on each student's learning progress and performance.
- **Multilingual Support:** Provide simplified course content in multiple regional and international languages.
- **Document Upload:** Allow students to upload PDFs, PPTs, notes, and textbooks for automatic simplification.
- **AI-Powered Question Generation:** Generate quizzes, MCQs, short-answer questions, and practice tests from the provided content.
- **Voice-Based Learning:** Add speech-to-text and text-to-speech features for hands-free learning.
- **Learning Progress Tracking:** Track topics studied, quiz scores, and improvement over time.
- **Advanced AI Agent:** Develop a more autonomous learning agent that can recommend what a student should study next based on their needs.
- **Mobile Application:** Extend the system into a mobile app for learning anytime and anywhere

Certificates Earned:

Credly Certificate(Getting Started with Artificial Intelligence)



Certificates Earned:

IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Deepak B

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Aug 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.